CHAINTRACE × MIDNIGHT

Private case attestations

Chain of custody for crypto investigations, without putting the investigation on a public chain.

GuTS805 · Midnight Buildathon, 2nd Wave

THE PROBLEM

A case report changes hands. Nobody can prove it wasn’t edited.

Handoffs and audits

Cases move between investigators, supervisors and courts. Each copy is a chance for findings to drift.

Public chains leak

Timestamping on a public ledger exposes suspect wallets, VASP attributions and who is investigating whom.

Plain hashes are weak

An unsalted hash can be guessed, and anyone can publish one. There is no rule about who may update it.

THE IDEA

Commit the hash. Keep the case.

Only a salted 32-byte commitment goes on Midnight. Only the holder of a private owner secret can update it.

HOW IT WORKS

Three steps, two people

01

Freeze

The case page exports canonical JSON, a random 32-byte salt and its SHA-256 commitment. The bundle stays on the device.

→

02

Attest

The DApp proves possession of the owner secret in zero knowledge and records the commitment through Lace.

→

03

Verify

A reviewer drops the bundle. Their browser re-hashes it and compares with the confirmed ledger. No wallet is needed.

DATA BOUNDARY

What the chain sees, and what it never sees

Data	Visibility	Where it lives
Case, findings, wallet addresses, officer	Private	Bundle file, shared only with chosen reviewers
32-byte salt	Private	Inside the bundle; blocks guessing the hash
Owner secret	Private	Browser only, sealed with PBKDF2 and AES-GCM
Commitment and revision	Public	Midnight ledger
Owner key (hash of secret)	Public	Midnight ledger, domain-separated

THE COMPACT CONTRACT

One case per deployment, three rules

Circuit	What it enforces
`constructor()`	Reads the owner secret from a private witness; publishes only its domain-separated hash
`attest(rev, c)`	Owner only; rev must equal the ledger revision (no stale or replayed transactions); no empty or duplicate commitment
`rotate_owner(k)`	Owner only; hands control to a new secret without revealing it; the old secret stops working

Compact 0.31.1 · compact-runtime 0.16.0 · ledger 8 (Preview, Preprod, Mainnet)

HONEST TRANSACTION STATES

No success message before the chain says so

Proving
Local proof server

Balancing
Lace pays fees and signs

Prepared
Proven, signed, not sent

Submitted
Transaction ID known

Confirmed
In a block and re-read

“Confirmed” appears only after the finalized transaction succeeded **and** a fresh ledger read shows the new commitment.

REVIEWER VERIFICATION

Every way a bundle can fail has a name

Result	Cause
`verified`	Bundle equals the confirmed on-chain commitment
`commitment-mismatch`	Case edited, or wrong contract
`bundle-invalid`	Wrong salt or damaged file
`superseded`	An older revision; the case has moved on
`contract-code-mismatch`	Address runs different circuits (verifier keys differ)

WHAT A MATCH MEANS

We claim integrity, not truth

Proves

The disclosed snapshot is byte-for-byte the version committed at that revision.

Whoever committed it held the case’s private owner secret.

Does not prove

That the findings or VASP attributions are correct.

Who the investigator is, or that evidence was authentic before it was committed.

TESTED, NOT JUST WRITTEN

Real proofs on a real local Midnight network

42

Unit, contract-simulator and UI tests: unauthorized calls, stale revisions, tampered bundles

6

End-to-end cases on node, indexer and proof server in Docker, through the DApp’s own code

1

Shared test vector, matched by the DApp, the Chaintrace frontend and a Python reference

STATUS AND NEXT WAVE

Live on Preview today

Built this wave

- Contract, DApp, Chaintrace integration
- Deployed and attested on Preview
- Contract 8f2b54a3…d11368, block 1,364,229
- Reviewer check verified; edit rejected

Next wave

- Selective disclosure of single findings
- Delegated reviewer access on chain
- Many cases in one contract

Prove the report.
Protect the case.

github.com/GuTS805/Chaintrace

Apache-2.0 · midnight/README.md has setup, tests and the demo walkthrough